

Leg pain

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I saw it once like two years ago... And that was it, I didn't need to see any more after that.

Kevin Ware, commenting on his gruesome complete tibial fracture that occurred while contesting a loose ball in the NCAA in 2013

The leg, defined as the anatomic region below the knee but above the ankle, is a common site of complaints among athletes, particularly distance runners. The term 'shin splints' is commonly used by runners as a non-specific reference to leg pain. The same term is also often used by health professionals to describe pain along the medial tibial border or to describe shin pain in general. Neither use of the term is pathologically precise. There are multiple causes with defined pathophysiology that should lead the clinician to a more specific diagnosis of leg pain in athletes. A more accurate and specific diagnosis allows for targeted treatment. Therefore, the term 'shin splints' should be abandoned in favour of a more specific, anatomic and diagnostic terminology.

This chapter focuses on four major pathologies that cause leg pain:

- medial tibial stress fracture
- anterior tibial cortical stress fracture
- medial tibial stress syndrome
- chronic exertional compartment syndromes.

CLINICAL APPROACH

Leg pain in athletes generally involves one or more of several pathological, anatomically specific processes.

1. **Bone stress.** A continuum of increased bone damage exists from bone strain to stress reaction and stress fracture.
2. **Vascular insufficiency.** This includes reduction in arterial inflow such as popliteal artery entrapment or vascular outflow due to venous insufficiency, thrombotic disease or vascular collapse due to elevated intracompartmental pressures.

3. **Inflammation.** This develops at the muscle insertions or along the tendons. Periosteal changes at the tibia posterior and soleus, and fascia to the medial border of the tibia may be due to traction or a variation of the stress injury to bone.
4. **Elevated intracompartmental pressure.** The lower leg has a number of muscle compartments, each enveloped by a thick inelastic fascia. The muscle compartments of the lower leg are shown in Figure 38.1. As a result of overuse or inflammation, these muscle compartments may become swollen and painful, particularly if there is excessive fibrosis of the fascia.
5. **Nerve entrapment.**

The clarification between these processes and the narrowing of the differential diagnosis begins with the

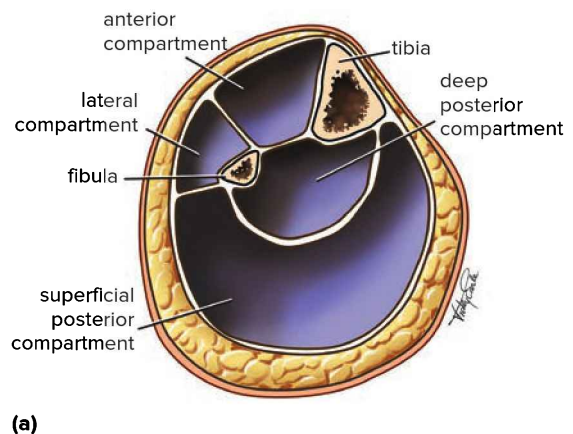


Figure 38.1 Cross-section of the lower leg (a) The various muscle compartments

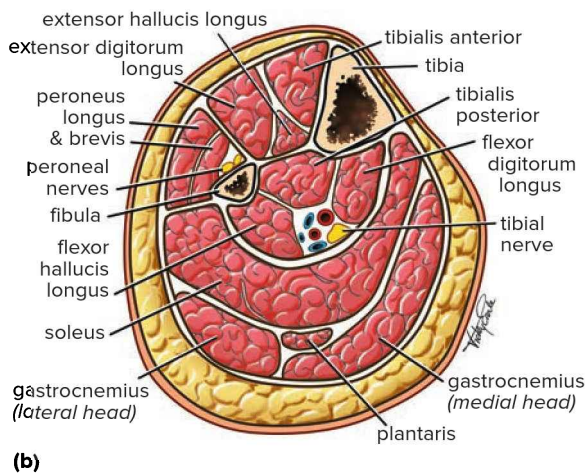


Figure 38.1 (cont.) (b) The individual muscles, nerves and vessels

history, narrowed by clinical examination findings, and confirmed with specific targeted imaging or clinical tests (Table 38.1). It is important to remember that two or three of these conditions may exist simultaneously. For instance, it is not uncommon to have a stress fracture develop in a patient with chronic periostitis. Periostitis or stress fracture may lead to intracompartmental swelling and tip a patient on the edge of symptomatic exertional compartment syndrome over the edge. This interrelationship is demonstrated in Figure 38.2. These co-existing conditions are usually managed differently and this explains why patients continue to have leg pain when only one condition has been addressed. The less common differential diagnoses for leg pain include tendinopathy, nerve entrapment, vascular claudication, neurogenic claudication, deep venous thrombosis (DVT), infection (osteomyelitis, cellulitis), metabolic bone disease, and tumours of bone and soft tissues, which may in rare cases affect prognosis and outcome of leg pain in athletes.

Role of biomechanics

Clinical experience suggests that abnormal biomechanics predisposes some individuals to pain on the anterior or medial border of the tibia (shin pain). Both extremes of foot type can contribute to the incidence of shin pain in athletes (Chapter 8). A rigid, cavus foot has limited shock absorption, thus increasing the impact pressure on the bone. In athletes with excessive pronation, the muscles of the superficial (soleus) and deep compartments (tibialis posterior, flexor hallucis longus [FHL], and flexor digitorum longus [FDL]) are placed at a relatively lengthened position and are required to contract eccentrically harder and longer to resist pronation after heel strike.

On toe off, these muscles must contract concentrically over a greater length to complete the transition to a supinated foot, creating a rigid lever for push off. With fatigue, these muscles fail to provide the normal degree of shock absorption. The chronic traction at the muscles' origins can lead to medial tibial stress syndrome. In chronic cases, this mechanism can contribute to the presentation of stress fractures or deep compartment syndromes.

The athlete with excessive pronation may also develop lateral shin pain or stress injuries of the fibula. Pronation of the fixed foot forces internal tibial rotation. With repetitive excessive pronation, the tibia and fibula are exposed to repetitive rotational (torque) stresses. These stresses are transferred across the fibula, tibia, and proximal and distal tibiofibular articulations. Based on these biomechanical stresses, overuse can lead to stress reactions or stress fracture, not only of the tibia but also of the fibula.

Motor imbalance can also lead to stress injuries in the lower extremity. Tight calf muscles, commonly the result of hard training, will restrict ankle dorsiflexion and increase the tendency for excessive pronation, leading to increased internal rotation of the tibia. Posterior tibial tendon weakness or deficiency can likewise contribute to foot pronation. Ankle instability secondary to chronic ankle sprains causes the athlete to overuse the peroneal tendons to compensate for ankle stability. This overuse can be just enough to send a athlete with borderline compartment syndrome over the brink to symptomatic complaints. To clearly understand biomechanics and have a foundation on which to create a differential diagnosis of leg pain, a complete knowledge of lower leg anatomy is essential (Figs. 38.1 and 38.3). When the presentation is atypical, one must expand the possibilities and consider a broad differential diagnosis for potential causes of leg pain in athletes (Table 38.2).

History

Taking a thorough history is essential for correct diagnosis of athletes with leg pain. Asking the right questions should narrow the diagnosis. Clarifying the pain complaints and assessing mechanism is key to the history. The history should be thorough and assess the presence of associated features with the pain such as paraesthesia or muscle hernia.¹ A complete history should include a broad review of systems to discover other facets such as metabolic issues, prior surgical procedures, developmental problems, contributing medical issues and social issues such as smoking. When the diagnosis does not classically fit the simple pain screen, a broader differential should be considered.

Table 38.1 The clinical characteristics and imaging features of common causes of shin pain in athletes

Condition	Pain	Effect of exercise	Associated features	Tenderness	Investigations
Bone stress reaction or stress fracture	Localised Acute or sharp Subcutaneous medial tibial surface or fibula	Constant or increasing Worse with impact	May be exacerbated by vibration with a tuning fork or therapeutic ultrasound (however, neither are a sensitive clinical test)	Subcutaneous medial tibial surface or fibula	Radiographs may be negative • Use magnified views • Look for callous or periosteal reaction MRI can stage severity and define prognosis but is also non-specific
Medial tibial stress syndrome	Diffuse pain on posteromedial border of tibia Variable intensity	Decreases as athlete warms up and stretches	Worse in the morning and after exercise Pes planus	Posteromedial edge of tibia at muscular insertions	Radiographs negative MRI shows diffuse oedema and periosteal thickening
Chronic exertional compartment syndrome	No pain at rest Ache, tightness, gradually building with exertion	Specific onset variable between athletes; usually 10–15 minutes into exercise Decreases with rest	Occasional muscle weakness or dysfunction with exercise Paraesthesia of nerve in affected compartment is possible	None at rest Anterior and lateral more common with exertion Occasionally related to palpable muscle herniation (superficial peroneal nerve)	Radiographs negative Bone scans negative Exertional compartment pressure testing is diagnostic Exertional MRI or infrared oxygen assessment may also be diagnostic
Popliteal artery entrapment	Pain in calf with exertion; not anterolateral 'Atypical compartment syndrome'	Worse with exertion especially active ankle plantar flexion	Pulses may be diminished with active plantar flexion (assessed by palpation or Doppler ultrasound)	Rarely in proximal calf	Radiographs negative MRI may reveal hypertrophic or abnormal insertion of medial gastrocnemius MR arteriography with provocative manoeuvres is diagnostic
Muscle-tendon injuries Strains Tendinopathy	Pain at pathologic site with resisted stretch	Pre-participation stretching usually helps	Short-term symptom relief with NSAIDs and ice	Pain can be at muscle belly, muscle-tendon junction, tendon or tendon insertion	Rarely required Radiographs usually negative MRI gives best view of soft tissue pathology

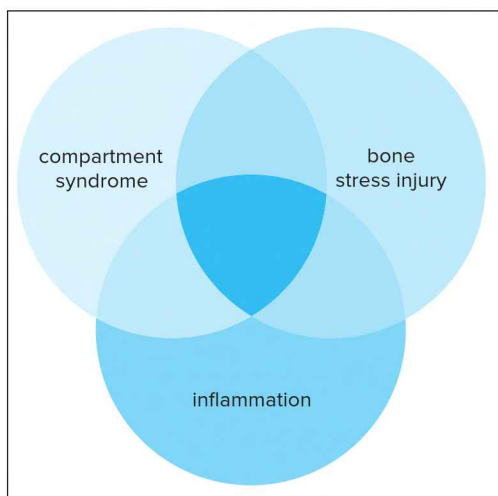
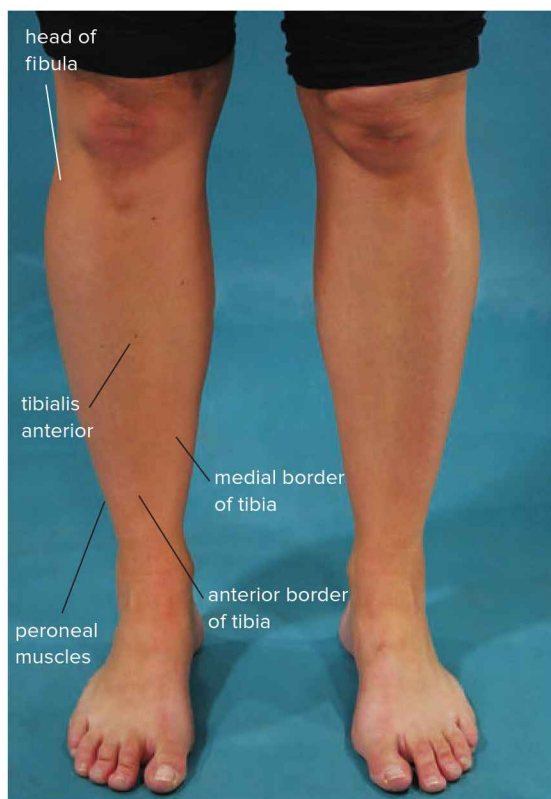


Figure 38.2 Possible interrelationship between the pathological sources of leg pain. These three entities need to be considered within a potentially critical factor—that of workload (represented by the surrounding box)



(a)

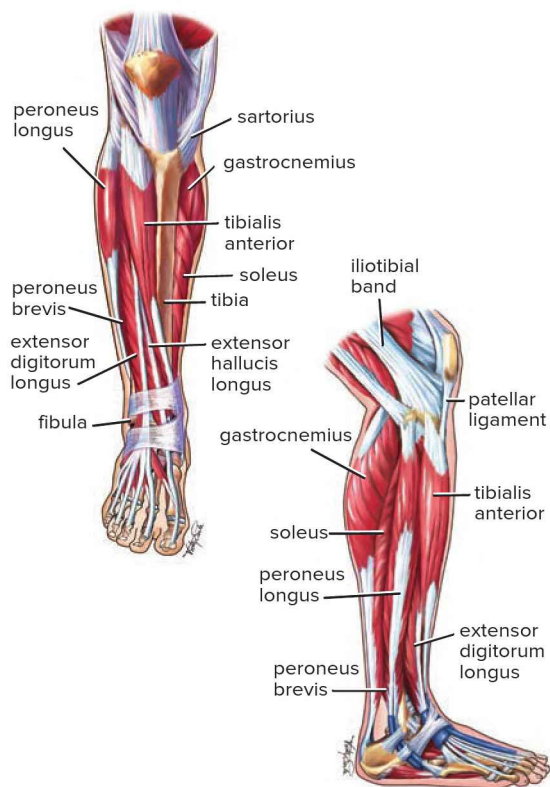
Figure 38.3 The leg (a) Surface anatomy (b) Anatomy of the lower leg from the front (left) and lateral aspect (right)

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Physical examination

In the examination of the patient with leg pain, it is important to palpate the site of maximal tenderness and assess the consistency of soft tissue. At rest, the physical examination for certain diagnoses (specifically chronic exertional compartment syndrome) is often unrewarding and the patient may be completely asymptomatic. The astute clinician will ask the athlete to reproduce the pain or symptoms via exertion or impact. This can be done on the athlete's playing surface, on a treadmill, or in and about the clinician's office by having the athlete run stairs or run around the block. A complete examination should be sequential, repeated the same way with each patient, and include observation, analysis of muscle function and range of motion, anatomically directed palpation, functional testing and diagnosis-specific testing (Fig. 38.4)

1. Observation. Assess lower limb alignment (varus/valgus, tibial torsion, pes planus) swelling, bruising, asymmetry:
 - a. standing (Fig. 38.4a)
 - b. walking—assess gait mechanics (forwards, backwards, on toes, on heels)
 - c. lying



(b)

Table 38.2 Causes of leg pain

Common	Less common	Not to be missed
Stress fracture • Medial tibia • Anterior tibia*	Stress fracture fibula	Tumours • Osteosarcoma • Osteoid osteoma
Acute fracture	Referred pain from spine	Infection (osteomyelitis, cellulitis)
Medial tibial stress syndrome	Chronic compartment syndrome • Superficial posterior	Acute compartment syndrome
Periosteal contusion	Vascular insufficiency/ Claudication	Chronic transition to acute compartment syndrome
Chronic compartment syndrome • Anterior • Lateral (peroneal) • Deep posterior	DVT (Chapter 39)	Chronic ankle injuries and Maisonneuve fracture
	Popliteal artery entrapment	Rare and unusual
	Femoral endarteritis	Syphilis
	Atherosclerotic disease	Sickle cell anaemia
	Superficial peroneal nerve entrapment	Hyperparathyroidism
	Muscle herniations	Sarcoidosis
	Baker's cysts or ganglion cysts	Rickets
	Osgood–Schlatter's disease (Chapter 44)	Paget's disease
	Pes anserine bursitis	Erythema nodosum
	Proximal tibiofibular subluxation	
	Electrolyte and metabolic dehydration cramping	

*Anterior tibial cortex stress fractures are less 'common' than medial tibial stress fractures but included here because of their clinical significance and because we discuss the two types of stress fractures in adjacent sections of this chapter.

2. Active movements. Assess motor function and range of motion:
 - a. plantarflexion/dorsiflexion—check pulses in full plantar or dorsiflexion; if diminished, consider popliteal artery entrapment
 - b. inversion/eversion
 - c. flexion and extension of the knee will put the gastrocnemius/soleus complex in a contracted or relaxed position
3. Passive movements. Assess true joint range of motion. May exacerbate pain in compartment syndromes:
 - a. plantarflexion (Fig. 38.4b)
 - b. dorsiflexion (Fig. 38.4c)
 - c. inversion/eversion
4. Resisted movements. Assess motor function. May exacerbate pain in muscle strains and tendinopathy:
 - a. plantarflexion/dorsiflexion
 - b. inversion/eversion
5. Functional tests:
 - a. hopping (Fig. 38.4d)—requires motor strength and landing skills. Rigid landing on heels exacerbates the pain of stress fracture
 - b. jumping
 - c. running—may bring on pain of exertional compartment syndrome or popliteal artery entrapment symptoms. Athletes should always reproduce symptoms
 - d. stair-climbing
6. Palpation. Evaluate pain distribution, warmth, swelling, pitting oedema, posterior cords or the presence of crepitus with motion:
 - a. tibia (Fig. 38.4e)—focal pain indicates stress fracture, diffuse pain over posterior medial border of tibia indicates medial tibial stress syndrome
 - b. fibula—the entire fibula should be palpated to identify focal pain related to a stress fracture. Severe eversion/external rotation ankle sprains may injure the syndesmotic connection between the tibia and fibula and be associated with a proximal fibula fracture (Maisonneuve fracture)
 - c. gastrocnemius, plantaris and soleus muscles (Fig. 38.4f)—look for muscle strains and ruptures. Tennis leg is a partial rupture of the medial head of the gastrocnemius vs plantaris
 - d. gastrocnemius-soleus aponeurosis (Fig. 38.4g)—assess for swelling and focal tenderness. May indicate tendinosis vs bursitis

Key diagnostic questions

Question	Clinical significance of response
<i>Was there an acute onset of pain?</i>	Fractures and tendon ruptures are usually acute traumatic events. In athletes, the acute onset of pain may be preceded by low-grade chronic pain of a stress fracture or tendinopathy.
<i>Is the pain chronic but stable?</i>	Pain that is getting worse over time should raise concerns of a tumour.
<i>Do you have a history of injury or prior leg pains?</i>	Old fractures or injuries can lead to scar tissue, stiffness and pain.
<i>Is the pain worse with impact?</i>	Stress fractures are normally exacerbated with impact. Medial tibial stress syndrome and muscle strains may also be made worse with loading and resistance.
<i>Is the pain worse with exertion?</i>	Pain absent at rest, that presents with exertion is classic for exertional compartment syndrome. Nonetheless, popliteal artery entrapment can have a similar presentation with posterior rather than anterior/lateral pain.
<i>Does the pain improve with warm-up and stretching?</i>	Medial tibial stress syndrome and muscle strains will frequently improve with pre-participation stretching, while stress fractures and exertional compartment syndrome generally do not.
<i>Does the pain get worse with stretching or resistance?</i>	Providing resistance to the muscle–tendon units including their origins and insertions should exacerbate symptoms related to medial tibial stress syndrome and muscle–tendon strains and tendinopathy.
<i>Where is the pain?</i> <i>Is the pain focal?</i> <i>Is the pain diffuse?</i>	The anatomic site of pain is the best physical clue to diagnosis. Focal pain over bone should raise suspicion of a stress fracture; focal pain over the muscle–tendon is likely a muscle strain or tendinopathy; diffuse pain over the posteromedial border of the tibia is likely medial tibial stress syndrome.
<i>Do you have swelling with the pain?</i> <i>Is it diffuse?</i> <i>Is it focal?</i>	Localised swelling is possible with a contusion, a stress fracture or muscle herniation. Diffuse swelling may indicate more significant injury or vascular problems like a DVT. Medial tibial stress syndrome may have palpable swelling about muscle insertions on the posterior medial border of the tibia.
<i>Do you feel electrical shooting pain?</i> <i>Do you have weakness with the pain?</i> <i>Do you get numbness with the pain?</i>	Electrical shooting pain, dermatomal loss of sensation and sclerotomal loss of motor power usually indicate nerve injury, entrapment or radiculopathy. Always check the lumbar spine.
<i>Does the pain get better with ice or non-steroidal anti-inflammatories (NSAIDs)?</i>	Pathologies associated with inflammation should improve with cold therapy and anti-inflammatories. Osteoid osteomas (a benign bone tumour) are known to have significant response to aspirin.
<i>Do you have pain at night?</i>	Pain that wakes a patient up at night should raise concern about tumours.

- e. superficial and deep posterior compartment (Fig. 38.4h)
- f. anterior and lateral compartment–post-exertion palpation may reveal tenseness in exertional compartment syndrome or palpable localised mass due to muscle herniation

7. Special physical examination tests:

- a. stress fracture test (Fig. 38.4i)–vibration may exacerbate pain associated with a stress fracture. Applying a vibrating tuning fork along the subcutaneous border of the tibia is a convenient and inexpensive test easily applied in the training room



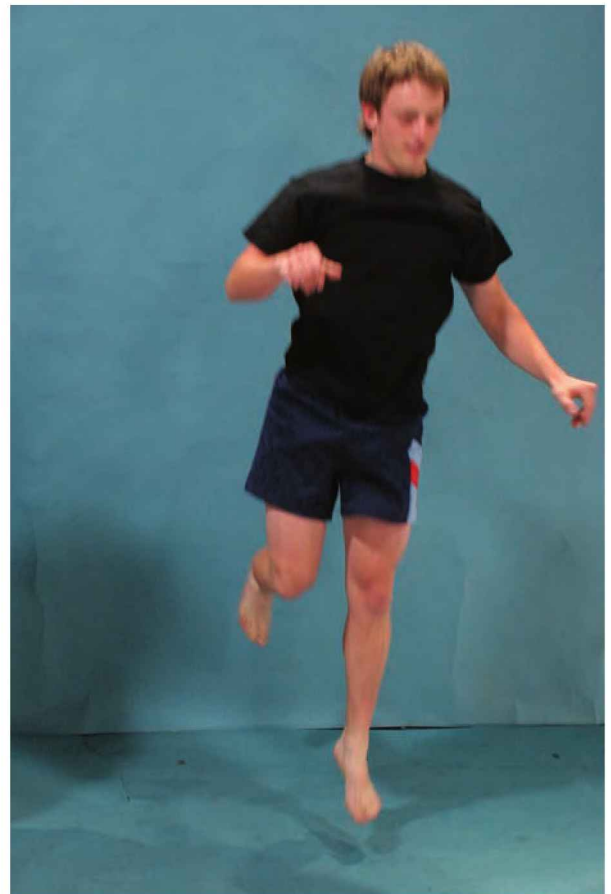
(a)



(c)



(b)



(d)

Figure 38.4 Examination of the patient with leg pain (a) Observation—standing. Assess lower limb alignment, swelling, bruising and any evidence of subperiosteal haematoma (b) Passive movement—plantarflexion. This may be restricted in anterior compartment syndrome (c) Passive movement—dorsiflexion. Measure the degree of passive dorsiflexion compared with the other side (d) Functional tests—hopping. If pain has not been reproduced, ask the patient to perform repeated movements, such as hopping, running or performing calf raises



(e)



(f)



(g)

Figure 38.4 (cont.) (e) Palpation—tibia. Palpate for the exact site of focal tenderness. Determine if pain is diffuse, focal or multifocal. Bony irregularity or subtle swelling may occur along the medial tibial border with medial tibial stress syndrome or stress fractures (f) Palpation—soleus muscle belly. A pincer grip is used (g) Palpation—soleus aponeurosis. Palpate for sites of tenderness and associated taut bands that may be a precipitating factor in the development of inflammatory shin pain

or office; however, this test's sensitivity (35–92) and specificity (19–83) is not well established²

- b. pain may also be exacerbated when therapeutic ultrasound is applied over a bone stress injury. However, this test has both poor sensitivity (64) and specificity (63),² and its use is not recommended.
- c. biomechanical examination (Chapter 8).

Key outcome measures

The medial tibial stress syndrome score³ is a valid and reliable outcome measure for medial tibial stress syndrome patients (Table 38.3). As there are few other tools to

monitor leg pain, this questionnaire could also be used for patients with lower leg stress fractures. There are currently no questionnaires to monitor chronic exertional compartment syndrome.

Investigations

Following the history and physical examination, extended work-up should target the most likely diagnosis. Imaging options included radiographs, MRI, CT scan, angiography, ultrasound and nuclear medicine scans. Routine radiography is rarely obviously positive in the diagnosis of leg pain. Careful inspection may reveal a subtle radiolucent line



(h)



(i)

Figure 38.4 (cont.) (h) Palpation—deep posterior compartment. This is palpated through the relaxed overlying muscles. In compartment pressure syndrome, the entire compartment feels tight in contrast to the localised tissue tightness of chronic muscle strain. Muscle or fascial hernias are occasionally found. The superficial posterior compartment should also be palpated (i) Special test—stress fracture. Applying vibration using a tuning fork over the site of tenderness can provoke pain in the presence of a stress fracture

Table 38.3 Patient-reported outcome measures for leg pain

Medial tibial stress syndrome score
• Simple self-administered questionnaire that evaluates pain and function in patients with medial tibial stress syndrome • 4 items • Score range 0–10 • Test-retest reliability (1 week): 0.82 • Construct validity: moderate to strong correlation with items of RAND-36 and with changes in the volume and intensity of patients' sporting activities • MDC: 0.69 • MCID: 0.69 • Available online: www.sportsoutcomemeasures.com

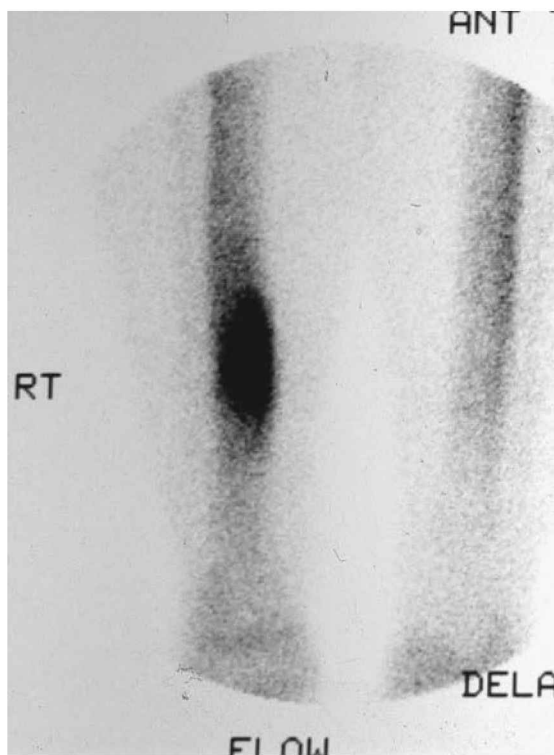
indicating a stress fracture, but radiography has a very low sensitivity for stress fractures generally and that applies particularly at this site (tibia).

Historically, radioisotopic bone scan was the next test in line to confirm the presence of a stress fracture or medial tibial stress syndrome. For stress fractures, a discrete focal area of increased uptake is seen on either the tibia (Fig. 38.5a) or fibula. In chronic medial tibial stress syndrome, the bone scan may show patchy areas of increased uptake along the medial border of the tibia. The absence of uptake does not preclude the diagnosis of medial tibial stress syndrome. No identifiable changes on the bone scan are associated with compartment syndrome.⁴

PRACTICE PEARL

MRI is the investigation of choice in patients with leg pain on the basis of its sensitivity to evaluate bony lesions, marrow changes and soft tissue injuries and correlate these findings with clinical symptoms.⁵

Typically, a stress fracture will appear on MRI as an area of periosteal oedema (Fig. 38.5b). The severity of injury documented on MRI (bone oedema only, unicortical radiolucent line or bicortical radiolucent line) has been correlated with healing time.^{6, 7} Medial tibial stress syndrome will appear with a broader area of oedema with thickening of the posterior medial periosteum or multiple small stress injuries of the bone. The MRI may also confirm the diagnosis of a muscle strain, muscle herniation, as well as benign (lipoma, cysts, osteoid osteoma) and malignant tumours. Magnetic resonance arteriography (MRA) with dynamic plantarflexion is the test of choice to confirm popliteal artery entrapment syndrome (Chapter 39).



(a)



(b)

Figure 38.5 Characteristic appearances on imaging (a) Bone scan of tibial stress fracture (b) MRI of tibial stress fracture showing fracture line (arrows) in the presence of marrow oedema

Investigation of chronic exertional compartment syndrome

For many years, it has been desirable to find a non-invasive method to confirm the diagnosis of chronic exertional compartment syndrome (CECS). Pre- and post-exertional MRI scans may reveal intracompartmental oedema confirming the diagnosis of an exertional compartment syndrome, but this is an expensive alternative. The role of MRI in the diagnosis of compartment syndromes is still unclear.^{5, 8, 9} Recently, diffusion tensor magnetic resonance imaging (DTI) and ultrasound have emerged as potentially attractive non-invasive alternatives.

Ultrasound provides a non-invasive, cost-effective and readily available tool to diagnose CECS of the anterior compartment of the lower extremity.¹⁰ Originally described by Gershuni et al.,¹¹ Martinson and Stokes¹² subsequently devised a technique using ultrasound to measure the thickness of the anterior compartment of the lower leg. Birtles et al.^{13, 14} expanded upon this technique to measure anterior compartment thickness in patients with CECS versus control subjects when performing eccentric and isometric exercises. Rajasekaran et al. measured anterior compartment thickness in patients at pre- and post-exertion periods, defined as running for at least 5 minutes up to 7 mph on a 20% incline on a treadmill. Patients with CECS had a significant difference and percentage change in anterior compartment thickness at all post-exertional time points versus resting levels when compared to control subjects.¹⁰

The use of ultrasonography may also contribute to understanding the pathophysiology behind CECS. In CECS patients, a convexity in the anterior fascial layer was noted during the post-exertional period. The combined effect of fascial bowing with a significant increase in anterior compartment thickness among CECS patients appears to offer an alternative explanation to the fascial non-compliance theory secondary to repetitive overuse and inflammation in the causation of CECS. In addition, recent studies show no significant difference in fascial elasticity in patients with CECS versus control subjects in the anterior leg compartment, which lie in stark contrast to previous findings.¹⁵

DTI is a non-invasive MRI marker that can serve as a diagnostic probe for pathologic conditions involving skeletal muscles. Ischaemia, inflammation, and injury to the muscle all provoke degenerative changes in the myofibril and are able to be detected via DTI metrics such as mean diffusivity and fractional anisotropy. Patients with defined CECS via T2-weighted imaging (>20% change in signal intensity in each compartment) had significantly increased diffusion rates that were anisotropic in nature after exercise with the most affected compartments being the anterior and superficial posterior compartment.

This was postulated to be from elevated temperatures increasing water diffusion rates, structural dilation of the myofibrils or interfibre oedema with exercise. Further development of this technique may lead to its use in the diagnosis and evaluation of response to treatment in CECS patients.^{16, 17}

PRACTICE PEARL

Intracompartment pressure measurement is the gold standard in confirming the diagnosis of chronic exertional compartment syndrome in the athlete.

Pre-exertion and immediate post-exertion measurements are essential to confirm the diagnosis. Devices used to measure the pressure have ranged from a wall blood pressure monitor with intravenous tubing and a three-way stopcock (Whiteside's technique), the transducer and pump used for an arterial line, laboratory systems and hand-held devices. In each case, the fascia is punctured percutaneously with a hollow bore needle, a needle with a side-port or by placement of a slit catheter.

A summary of the recommended techniques with appropriate exercises to produce pain is shown in Table 38.4.

The anterior compartment is relatively easy to find just lateral to the anterior border of the tibia with the lateral compartment being just lateral to the anterior compartment and anterior to the fibula. The deep posterior compartment or, if present, the tibialis posterior compartment may be more difficult to locate but is usually accessed posterior to the posteromedial edge of the tibia. Some experts recommend the use of ultrasound guidance to ensure correct catheter placement.¹⁸

PRACTICE PEARL

In clinical practice, the most widely used diagnostic criteria has been a basal intracompartmental pressure (ICP) of >15 mmHg, 1 minute post-exertion >30 mmHg and a 5 minute post-exertion >20 mmHg.¹⁹

There is currently no universally accepted method or ICP threshold for the diagnosis of CECS. Pressures between 10–20 mmHg or those that decrease with exercise are considered to be inconclusive. A systematic review showed the strength of evidence supporting these criteria to be weak, owing to the fact that there is high variability and inconsistency in pressure measurements, operator experience, catheter type used, volume of

Table 38.4 Compartment pressure testing

Compartment	Location of catheter	Exacerbating exercise	Compartment pressures (N=10 mmHg, post-exercise 5 min)
Deep posterior	Junction of lower and middle third of tibia Aim deep posteriorly just behind the posteromedial tibial border	Treadmill/running Stairs Run/jump Pulleys in PF/DF Repeated case raises Isokinetic PF with IV/EV Sports-specific challenges	>25 mmHg post exercise or an increase of >10 mmHg compared to resting baseline
Superficial posterior	Aim more posteriorly from deep posterior entry into medial gastrocnemius or soleus	Treadmill Stairs Repeated calf raises	Same as above
Anterior	Mid-belly of tibialis anterior Anterior to intermuscular septum (half-way between fibula and anterior border of tibia)	Repeated DF Treadmill/running Stairs Sports-specific challenges	Same as above
Lateral (Peroneal)	Mid-belly of peroneals Posterior to intermuscular septum	Repeated IV/EV Treadmill/running Stairs	Same as above

Ankle movements: PF = plantarflexion, DF = dorsiflexion, IV = inversion, EV = eversion

instilled fluid used and depth of catheter placement.²⁰ Admittedly, it is still the most common criteria used as there is no established technique that has been shown to be more effective.

Dynamic measurement of ICP has been introduced to address shortcomings associated with current static time driven methods of ICP measurement. This includes avoidance of multiple needle sticks and data collection that is reliable, extensive and not prone to artefact and technical issues associated with fluid-filled catheters/devices. Dynamic measurement allows changes in ICP to be matched to symptom development in real time during a standardised exercise challenge. This offers a more sensitive and specific method to diagnose CECS. Compared to the widely used Pedowitz criteria of ICP >30 mmHg 1 minute post exercise and ICP >20 mmHg 5 minutes post exercise, a threshold level of 105 mmHg achieved during active exercise was found to provide the greatest diagnostic validity in patients exhibiting symptoms of CECS. This indicates that continuous measurement of ICP throughout the duration of exercise improves its diagnostic utility in accurately diagnosing CECS.²¹

By increasing the diagnostic validity of methods used to diagnose CECS, we can potentially avoid poor outcomes after fasciotomy through better patient selection. By being able to differentiate between patients who exhibit symptoms of CECS but ultimately lack a pressure-related aetiology versus those that have a pressure-related condition, we can identify potential patients who stand to have a higher likelihood of a positive outcome after fasciotomy. Other authors have promoted the use of non-invasive methods such as near-infrared spectroscopy, and results are promising but require further validation.⁹

Additional work-up for more atypical causes of leg pain include:

- EMG/nerve conduction studies
 - peripheral nerve entrapments & metabolic neuropathy
- Ankle/brachial index
 - vascular claudication
- Venous Doppler
 - deep venous thrombosis
- Laboratory
 - CBC (complete blood count) with differential—infection or osteomyelitis
 - ESR (erythrocyte sedimentary rate)—inflammation, rheumatologic conditions
 - sickle cell preparation—sickle cell anemia
 - urine analysis with uromyoglobin—rhabdomyolysis
 - CPK (creatinine phosphokinase), myoglobin—rhabdomyolysis, myopathy
 - PT (prothrombin time), APTT (activated partial thromboplastin time)—deep venous thrombosis

- D-dimer—deep venous thrombosis
- metabolic panel—hypokalemia, hypocalcemia, hypomagnesium, etc
- T₃ (triiodothyronine), T₄ (thyroxine), TSH (thyroid stimulating hormone)—thyroid myopathy.

MEDIAL STRESS FRACTURE OF THE TIBIA

Stress fractures are more commonly a cause of leg pain in athletes in impact, running and jumping sports. Overall limb and foot alignment as well as limb length discrepancy may also play a role. Runners with stress fractures have been shown to have a reduced cortical strength, a reduced cross-sectional area of cortical bone (thinner cortices) and reduced muscle cross-sectional area compared to athletes without a history of stress fractures.²² The incidence of stress fractures increases on more rigid, unforgiving surfaces.

PRACTICE PEARL

Approximately 90% of tibial stress fractures will affect the posteromedial aspect of the tibia. The most common site is near the junction of the middle and distal thirds.

Proximal metaphyseal stress fractures may lead to more time loss from sport, as they do not respond as well to the functional bracing that allows earlier return to play. We discuss the dreaded 'anterior cortical tibial stress fracture' in a separate section.

Assessment

A typical posteromedial tibial stress fracture presents with the following clinical features:

- pain is gradual in its onset, aggravated by exercise, often with a recent history of an increase in training intensity
- pain may occur with walking, at rest or even at night.
- examination shows localised tenderness over the tibia.
- biomechanical examination may show a rigid, cavus foot incapable of absorbing load, an excessively pronating foot causing excessive muscle fatigue or a leg-length discrepancy
- tenderness to palpation along the medial border with obvious tenderness. Note that stress fractures of the posterior cortex occasionally produce symptoms of calf pain (Chapter 39) rather than leg pain
- bone scan and MRI appearances are of a stress fracture of the tibia, as shown in Figures 38.5a and 38.5b. MRI scans are particularly valuable, as the extent of oedema

and cortical involvement has been directly correlated with the expected return to sport⁷

- CT scan may also demonstrate a stress fracture (Fig. 38.6).

Treatment

Prior to initiating treatment or during the treatment plan, it is important to identify which factors precipitated the stress fracture. The most common cause is an acute or chronic change in training habits, such as rapidly increasing distance over a short period,²³ beginning double-practice days after a period of reduced training or changing to a more rigid playing surface. Footwear, abnormal biomechanics, and repetitive impact sports such as running and gymnastics have also been implicated. Systematic reviews have shown that the use of custom semi-rigid orthotics led to lower incidence of stress fractures in athletes and military recruits.²⁴ Coaches can play a key role in modifying training patterns to reduce the risk of these injuries.

In women, reduced bone density may increase risk of stress fracture.²⁵ All female athletes with a first time stress fracture should be screened for Relative Energy Deficiency in Sport (RED-S) (Chapter 4). Research has focused on reducing the vertical loading rate and ground reactive force by modifying stride length, gait retraining and running speed.²⁵⁻²⁹ These studies suggest that a softer running style can prevent the development of stress fractures. This can be attained through:

1. a shorter stride length
2. slower running speeds
3. reduced mileage (reduces the number of impacts).

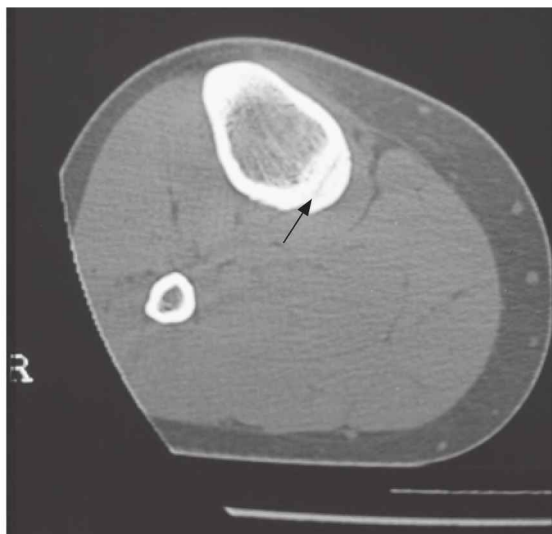


Figure 38.6 Axial CT at the region of the mid-calf showing the posteromedial aspect of the right tibia affected by a longitudinal stress fracture

The classic treatment plan is as follows:

- initial period of rest (sometimes requiring a period of non weight-bearing on crutches for pain relief) until the pain settles
- use of a pneumatic brace (Air-Stirrup® Leg Brace, Aircast®, New Jersey, USA), which has markedly reduced return to activity time compared with 'standard' recovery times in two of three studies^{30, 31} and compared with a 'traditional treatment' group in the third.³² In this latter study, the brace group returned to full, unrestricted activity in an average of 21 days compared with 77 days in the traditional group. The brace should extend to the knee as the mid-leg version may actually increase the stress across a midshaft stress fracture. Routine use of this treatment modality, implies rest and full-time use for 1-2 weeks. During that time, the athlete should notice excellent pain control while in the brace. If this positive response is achieved, then the athlete is allowed to gradually return to impact activities and play in the brace. If pain returns, the period of rest is extended. As long as the pain is well controlled, the athlete is allowed to return to full athletic participation in the brace. Most athletes will complete their competitive season using the brace as complete healing may not occur while the athlete is still loading the leg with or without the brace. Using this plan, there have been no reported cases of progression to complete catastrophic fracture of the tibia
- if pain persists, continue to rest from sporting activity until the bony tenderness disappears (4-8 weeks)
- once the patient is pain-free when walking and has no bony tenderness, gradually progress the quality and quantity of the exercise over the following month. Please note, the athlete is asked to continue to use a pneumatic brace to complete the current season until an appropriate period (4-8 weeks) of rest can occur
- cross-training with low-impact exercises including swimming, cycling, and water running maintains conditioning and reduces risk of recurrence
- pain associated with soft tissues distal to the fracture site can be treated using soft tissue techniques
- general principles of return to activity following overuse injury should be followed (Chapter 18).

Prevention of recurrence

- Determine whether excessive training and abnormal biomechanics precipitated the stress fracture.
- Inadequate calorie balance is a risk factor for stress fractures (Chapter 4).
- Alternative treatments including electrical stimulation and low-intensity pulsed ultrasound (LIPUS)^{33, 34} have

been tried as a way to accelerate healing or prevent recurrence but have not proven effective (Chapter 17).

- Bisphosphonates have no place in the treatment or prevention of these stress fractures.³⁵

STRESS FRACTURE OF THE ANTERIOR CORTEX OF THE TIBIA

PRACTICE PEARL

Stress fractures of the anterior cortex of the mid-shaft of the tibia need to be considered separately because they are prone to delayed union, non-union and complete fracture.

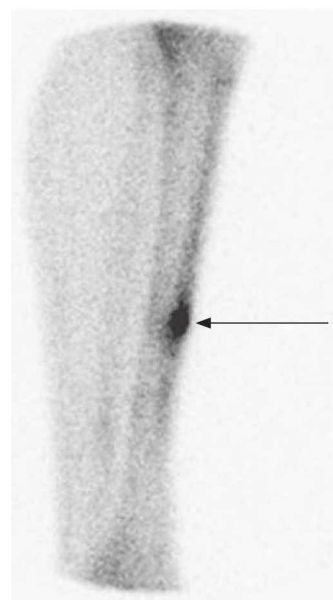
Stress fracture of the anterior cortex of the medial third of the tibia presents with diffuse dull pain aggravated by physical activity. The bone is tender to palpation at the site of the fracture and periosteal thickening as evidenced by a palpable lump may be present if the symptoms have been present for some months. Isotopic bone scan (Fig. 38.7a) shows a discrete focal area of increased activity in the anterior cortex. The radiographic appearance at this stage shows a defect in the anterior cortex, which is termed the 'dreaded black line' (Fig. 38.7b). This appearance is due to bony resorption and is indicative of non-union.

The mid-anterior cortex of the tibia is vulnerable to non-union for two reasons: it has a relatively poor blood supply and it is an area under tension due to the morphologic bowing of the tibia. Excessive anterior tibial bowing is often noted in association with this fracture. In general, the prognosis for these resistant stress injuries is guarded, with an elevated risk of delayed or non-union. One study presented some success with the use of pneumatic air braces in this population, avoiding the need for surgery.³⁶ The average return to unrestricted activities was 12 months with this form of treatment. Other options for treatment include pulsed electromagnetic stimulation, surgical excision and bone grafting and transverse drilling at the fracture site. Chang and Harris described five cases treated with intramedullary tibial nailing.³⁷ They had two excellent and three good results.

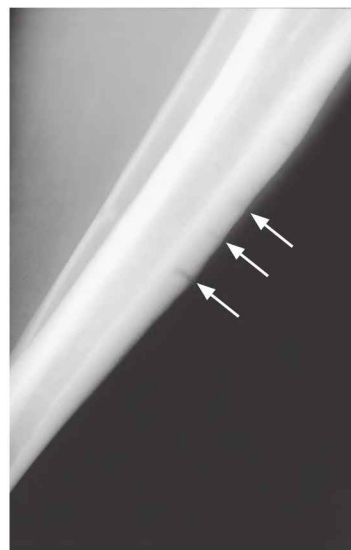
Treatment

Our current treatment protocol is as follows:

- immediate application of a pneumatic brace
- discontinuing anti-inflammatory medications and smoking
- thorough screening for associated nutritional, bio-mechanical and nutritional risk factors
- for all elite-level athletes, application of electrical or ultrasonic bone stimulation
- if no progress by 4-6 months, intramedullary rodding (+/-) bone grafting, debridement or drilling is recommended.



(a)



(b)

Figure 38.7 Stress fracture of the anterior tibial cortex (a) Bone scan appearance: lateral view highlights the anterior location of increased radioisotope uptake (b) Plain radiograph appearance of multiple 'dreaded black lines'

MEDIAL TIBIAL STRESS SYNDROME

As noted previously, there has been a tendency in the past to categorise all shin pain, especially that which is not a stress fracture, under the term 'shin splints'.³⁸ Indeed, 'shin splints' is more of a vague symptom athletes describe for leg pain most commonly along the posterior medial border of their tibia. Imaging techniques such as CT and MRI allow us to make more precise anatomical and pathological diagnoses of patients with leg pain.

PRACTICE PEARL

The most descriptive term that accounts for the inflammatory, traction phenomena on the medial aspect of the leg more common in runners is medial tibial stress syndrome or medial tibial traction stress syndrome.

However, recent studies have argued that the process may be more similar to stress injury of the bone and argue that no inflammatory process is present.

The patient with medial tibial stress syndrome complains of diffuse pain along the medial border of the tibia (the junction of the lower third and upper two-thirds of the tibia), which usually decreases with warming up. The incidence has been reported between 4% and 35% in military personnel and athletes.^{39, 40} More focal pain should alert the examiner to the possibility of a true stress fracture. The athlete can often complete the training session but pain gradually recurs after exercise and is worse the following morning. Historically, the tibialis posterior was thought to be the source of the pain, but more recently the soleus and flexor digitorum longus have been implicated.³⁹

Risk factors

A number of factors may contribute to the increased stress and traction on the posterior medial aspect of the tibia. These include excessive pronation (Chapter 8), training errors, shoe design, surface type, muscle dysfunction, fatigue and decreased flexibility.⁴¹ Other risk factors that have been reported include female sex, higher body mass index, greater internal and external rotation of the hip, increased calf girth, and a history of previous stress fractures or use of orthotics.^{40, 42}

The biomechanics of medial tibial stress syndrome relates to the sequence of events that occurs with walking and running.⁴³ During midstance, foot pronation provides shock absorption and an accommodation to the varied

terrain. The medial soleus is the strongest plantar flexor and invertor of the foot. The soleus muscle eccentrically contracts to resist pronation. Excessive pronation due to pes planus or overuse combined with repetitive impact loading leads to chronic traction over its insertion onto the periosteum on posterior medial border of the tibia leading directly to medial tibial traction stress syndrome and medial tibial stress syndrome.

Metabolic bone health may also contribute. Magnusson et al. found athletes with medial tibial stress syndrome had lower bone mineral density at the affected region compared with control and athletic control subjects.⁴⁴ Bone mineral density was also decreased on the unaffected side in subjects with unilateral symptoms.⁴⁴ These athletes regained normal bone mineral density after recovery from their symptoms.⁴⁵ Reduced bone density or bone load tolerance may contribute to the increased risk of medial tibial stress syndrome seen in female military recruits. A study examining possible risk factors for the development of medial shin pain in military recruits showed that females were three times as likely to develop the syndrome.⁴⁶ Beyond sex, no other risk factors of statistical significance were noted; however, increased hip range of motion (both internal and external rotation) and lower lean calf girth were associated with medial shin pain in the male recruits.⁴⁶

Radiographs are normally negative with medial tibial stress syndrome. However, it is sometimes possible to see periosteal reactions or localised swelling. Isotopic bone scan may show patchy, diffuse areas of increased uptake along the medial border of the tibia. This is in contrast to stress fractures, which should show focal uptake. In the early stages of medial tibial stress syndrome, however, the bone scan appearance may also be normal. MRI was found to have similar sensitivity and specificity to isotope bone scan.⁹ Interestingly, there were a number of abnormal bone scan and MRI appearances in the asymptomatic control group in that study.

Treatment

Most athletes will present with a long history of complaints, having tried a number of home remedies, stretches, medicines or cold treatment. Assessing previous treatments in terms of what provided relief and what exacerbated the problem is beneficial. While heat or whirlpool may improve flexibility and warm up muscles, it also increases the circulation to the region, which can increase symptoms.

The foundation of treatment is based on symptomatic relief, identification of risk factors and treating the underlying pathology. Symptomatic treatment begins

with rest, ice and analgesia as needed. Switching to pain-free cross-training activities such as swimming or cycling can keep the athlete active. Craig has suggested that no current method of treatment is better than rest alone; yet she still suggests the use of shock-absorptive insoles as promising.^{39, 47}

In resistant cases, immobilisation and protected weight-bearing may be necessary to rest the chronic tension placed on the soleus insertion with repeated weight-bearing. Taping techniques are only effective if they control foot pronation.

A critical facet of treatment is based on a careful assessment of foot alignment and gait mechanics. Permanent relief can occasionally be achieved through appropriate footwear and the application of cushioned orthotics (for shock absorption assistance) with a semi-rigid medial arch support (to support the pronated foot). We have found positive results in our patients, at least in the initial phase, by treating them with the same knee-high pneumatic splint that we use for stress fractures (Air-Stirrup Leg Brace®, Aircast®, New Jersey, USA).

Alternative modalities can be effective in relieving pain and should be considered in patients with medial tibial stress syndrome. The entire calf muscle should be assessed for areas of tightness or focal thickening that can be treated with appropriate soft tissue techniques (Chapter 17). Digital ischaemic pressure may be applied to the thickened muscle fibres of the soleus, flexor digitorum longus and tibialis posterior adjacent to their bony attachment, avoiding the site of periosteal attachment, which may prove too painful (Fig. 38.8a). The effect may be enhanced by adding passive dorsiflexion and plantarflexion while digital ischaemic pressure is applied. Transverse frictions are commonly used in clinical practice to treat focal regions of muscle thickening in the soleus and flexor digitorum longus.

Abnormalities of the tibialis posterior may be treated through the relaxed overlying muscles. Sustained myofascial tension can be applied parallel to the tibial border, releasing the flexor digitorum longus and along the soleus aponeurosis in the direction of normal stress with combined active ankle dorsiflexion (Fig. 38.8b).

Physical therapy programs have focused on motor strengthening and flexibility, especially proprioceptive neuromuscular facilitation (PNF) stretching. In addition, electrical stimulation, iontophoresis and ultrasound have been attempted, with mixed results. Prolotherapy injection with agents intended to accelerate the healing process and platelet-rich plasma injections have also been performed, but very little quality research is available to validate their efficacy. In resistant cases, surgical release (with or without periosteal tissue resection or ablation)



(a)



(b)

Figure 38.8 Soft tissue therapy in the treatment of shin pain (a) Digital ischaemic pressure to the medial soleus aponeurosis and flexor digitorum longus. This can be performed with passive and active dorsiflexion (b) Sustained myofascial tension along the soleus aponeurosis in the direction of normal stress combined with active ankle dorsiflexion

of the superficial and posterior compartments from its conjoined insertion onto the posteromedial border of the tibia can be performed with a projected success rate of 70% improvement in high-performance elite athletes.^{38, 48}

CHRONIC EXERTIONAL COMPARTMENT SYNDROME

Compartment syndrome is defined as increased pressure within a closed fibro-osseous space causing reduced blood flow and reduced tissue perfusion subsequently leading to ischaemic pain and possible permanent damage to the tissues of the compartment.⁴⁹ This may be acute, chronic (exertional) or convert from chronic to acute. Together with stress fractures and medial tibial stress syndrome, CECS is a key differential diagnosis of leg pain in athletes, especially in distance runners and other endurance athletes. The syndrome is frequently bilateral. When the pain is in the calf, the clinician should also consider popliteal artery entrapment syndrome (Chapter 39).

Pathogenesis

While classically, exertional compartment syndrome was believed to be an ischaemic phenomenon like acute compartment syndrome, the exact aetiology of CECS is still unclear. Repetitive overuse followed by associated inflammation may lead to fibrosis and therefore reduced elasticity of the fascia surrounding the muscle compartments. As a result, when the patient exercises, the muscles attempt to expand but are unable to do so.

Biopsies have revealed abnormally thickened, non-compliant fascia. A series of biopsies at the fascial-periosteal interface revealed varying degrees of fibrocytic activity, chronic inflammatory cells, vascular proliferation and a decrease in collagen irregularity, which suggests attempted remodelling was taking place.⁵⁰ As a result of this stiffened, abnormal fascial compartment, when the patient exercises, the muscle attempts to expand, but is resisted by a less compliant fascia. This results in increased pressure and subsequent pain. As noted previously, ischaemia likely plays some role; however, this has not been well documented.

Within a tight fascial compartment, it is possible that normal metabolic activity during exercise leads to a sufficient increase in pressure to compromise tissue perfusion. Birtles et al. induced similar symptoms to those of compartment syndrome by restricting venous flow during exercise.⁵¹ Edmundsson et al. noted that when patients with CECS had muscle biopsies at the time of their fascial release, laboratory analysis revealed lower capillary density, lower number of capillaries around muscle fibres and lower density of capillaries per muscle fibre area.⁵²⁻⁵⁴ Thus, the reduced

microcirculation likely contributed to the development or was secondary to the CECS.

Typical clinical features of CECS are the absence of pain at rest with increasing ache or pain and a sensation of tightness with exertion.

Symptoms usually resolve or significantly dissipate within several minutes of rest. Rarely, athletes will develop paraesthesia or motor weakness with exertion. At rest, physical examination is usually unremarkable. With exertion, the compartment may be tense on palpation, and it may be possible to see bulges or herniations in the leg.

The most common compartment involved is the anterior compartment, which presents as anterolateral pain with exertion. The other two common compartments are the lateral compartment, which may include paraesthesias in the distribution of the superficial peroneal nerve to the dorsum of the foot, and the deep posterior compartment, which is usually associated with posteromedial tibial pain. Symptoms arising from the superficial posterior compartment are rare.

Clinical features

Work-up and screening should always include an assessment of limb and foot alignment, evaluation of the biomechanical demands of the specific sport including training surface and shoe-wear, a history of previous injuries or trauma, and a screen for overlapping pathology such as stress fractures and medial tibial stress syndrome, as well as metabolic and nutritional factors. In one study, diabetes mellitus was implicated as a risk factor for developing CECS.⁵⁵ Radiographs are frequently obtained as an inexpensive screening tool for associated bone pathology.

The definitive diagnosis is made on the basis of intracompartmental pressure measurements (Table 38.4). The use of near-infrared spectroscopy has shown promise as a non-invasive alternative, but is expensive and has not yet become commonly used.⁹ A study comparing near-infrared spectroscopy to MRI and intracompartmental pressure measurements found that the sensitivity of near-infrared spectroscopy (85%) was superior to either MRI or intracompartmental pressure measurements (both 77%).⁹

More recently, Williams and associates have suggested that non-painful neurosensory testing can be performed using a pressure-specified sensory device pre- and post-exertion, which may guide the clinician both in diagnosing CECS and in assessing the efficacy of treatment.⁵⁶ Other investigators have looked more deeply into advanced imaging techniques to assist with a non-invasive diagnosis of CECS, including ultrasound, Tc-99-tetrofosmin single-photon emission CT, a novel dual birdcage coil, in-scanner MRI protocols and DTI.^{57, 58} Nonetheless, intracompartmental pressure measurements remains the gold standard.

Deep posterior compartment syndrome

Deep posterior compartment syndrome typically presents as an ache in the region of the medial border of the tibia or as chronic calf pain. Beware the multiple other causes of calf pain including popliteal artery entrapment syndrome (Chapter 39). The deep posterior compartment contains the flexor hallucis longus, flexor digitorum longus and tibialis posterior (Fig. 38.1). Occasionally, a separate fascial sheath surrounds the tibialis posterior muscle, forming an extra compartment that may provoke symptoms independent of the other compartments.

Active, passive or resisted motion of these muscles may exacerbate pain. The patient describes a feeling of tightness or a bursting sensation that increases with exercise. Distal symptoms such as weakness or pins and needles on the plantar aspect of the foot may indicate tibial nerve compression. Small muscle hernias occasionally occur along the medial or anterior borders of the tibia after exercise.

On examination, there may be tenderness along the medial aspect of the tibia, but this is often relatively mild. Due to the deep nature of the compartment, palpable fascial tightness is less obvious in comparison to anterior or lateral compartment syndromes. Nonetheless, the experienced clinician may be able to discern the difference between palpable tightness in the deep compartment and fascial thickening and induration found in association with medial tibial stress syndrome.

To measure deep posterior compartment pressures, the needle or catheter is inserted from the medial aspect through two layers of fascia aiming posterior to the tibia (Fig. 38.9). A range of exercises can be used to exacerbate complaints, such as running, jumping, stair climbing, repeated calf raises or resisted plantarflexion and dorsiflexion using elastics, pulleys or isokinetic resistance machines. Routinely, we ask patients to run 5 minutes into their pain to ensure a valid test. It is important to reproduce the patient's pain, otherwise the test is not considered valid.

Post-exertional measurements must be obtained immediately after ceasing exercise and may be repeated again after 10 minutes. Normal compartment pressures are regarded as being between 0 and 10 mmHg. Maximal pressure during exercise of > 25–30 mmHg, an elevation of pressures > 10 mmHg or a resting post-exercise pressure > 25 mmHg is necessary for the diagnosis of chronic compartment syndrome (Table 38.4). If the elevated pressure takes more than 5 minutes to return to normal, this may also be significant.

Treatment

Treatment of isolated deep posterior exertional compartment syndrome usually begins with a conservative regimen of reduced exercise and deep massage therapy.



Figure 38.9 Compartment pressure testing—deep posterior compartment. The Stryker catheter is inserted into the deep posterior compartment

All contributing factors and overlapping diagnoses must be carefully considered. Effective soft tissue techniques include longitudinal release with passive and active dorsiflexion (Fig. 38.10) and transverse frictions. Dry needling of the deep muscles may also be helpful. Assessment and correction of any biomechanical abnormalities, especially excessive pronation, must be included.

Isolated deep posterior exertional compartment syndrome is uncommon and may be confused with medial tibial stress syndrome, popliteal artery entrapment syndrome, vascular claudication and stress fractures. Indeed, it is not surprising that initial treatment is the same as that for medial tibial stress syndrome. Unfortunately, if associated diagnoses or contributing factors cannot be identified and if pressures are elevated, symptoms are usually refractory to treatment. In this case, surgical release is necessary.

The surgical approach is along the posterior medial edge of the tibia and may be performed through a single or two small incisions. The saphenous vein lies directly along the path to the fascial insertion onto the posteromedial border of the tibia. Extreme care must be used to control all bleeding at the time of surgery as injury to one of the branches is common and increases the risk of postoperative haematoma or cellulitis.⁵⁹

Some authors have suggested a benefit of fasciectomy (removal of a portion of fascial tissue) over fasciotomy (simple incision) due to concerns that the fascial insertion and sheath reforms.¹⁸ They argue that this periosteal stripping serves an added role of treating any associated medial tibial stress syndrome and it assures release of any anatomic variations of tibialis posterior compartments. Due to the extensive nature of the procedure, which requires longer incisions and has increased risk of

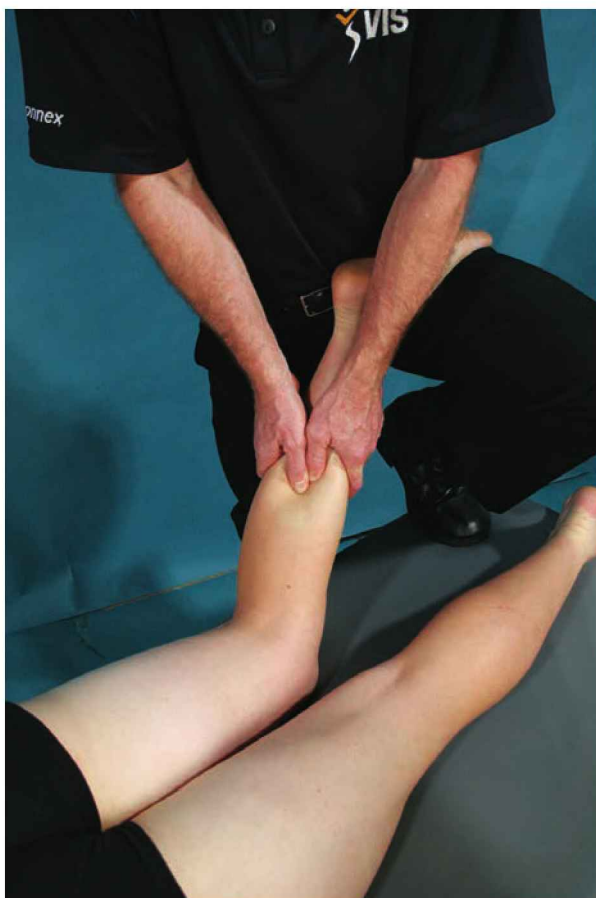


Figure 38.10 Soft tissue therapy in the treatment of posterior compartment syndrome—longitudinal release with active or passive dorsiflexion

complications, we recommend this extensive approach only in revision cases. In addition, we believe it is prudent to limit treatment to the most affected compartment in patients who have positive anterior or lateral compartment pressures, but only borderline pressures in the deep compartment. This may reduce the risk of postsurgical complications, which is elevated when releasing the posterior compartments.⁵⁹⁻⁶¹

Anterior and lateral exertional compartment syndromes

The anterior compartment contains the tibialis anterior, extensor digitorum longus, extensor hallucis longus and peroneus tertius muscles, as well as the deep peroneal nerve. The lateral (peroneal) compartment contains the peroneus longus and brevis tendons as well as the superficial peroneal nerve. For anterior compartment pathology, exertional pain occurs just lateral to the anterior border of the shin. Paraesthesia may also occur in the first web space. For lateral

compartment pathology, pain is felt just anterior to the fibula and paraesthesia may occur over the dorsum of the foot. The intermuscular septum (raphe) between the two compartments can be seen in thin individuals by looking for the indentation of skin when you squeeze the soft tissues between the anterior border of the tibia and fibula.

Clinical examination at rest will usually be normal or there may be palpable generalised tightness of the anterior or lateral compartment with focal regions of excessive muscle thickening. It is also important to assess the plantarflexors, especially the soleus and gastrocnemius. If these antagonists are tight, they may predispose to anterior compartment syndrome. Muscle herniation may be palpable with exertion most commonly occurring 5-7 cm proximally to the distal tip of the fibula where the superficial peroneal nerve penetrates through the lateral compartment fascia. Diagnosis of anterior and lateral exertional compartment syndrome is confirmed with pre- and post-exertional compartment testing (Table 38.4).

Treatment

Treatment is based on the same principles as for the deep posterior compartment. All contributing factors should be assessed and treated. Lowering the heel in the athlete's shoe or orthotics may reduce the load of the anterior muscles and alleviate pain. Soft tissue techniques, such as sustained myofascial tension combined with passive and active plantarflexion (Fig. 38.11), transverse frictions and dry needling may alleviate symptoms.

Dr Andrew Franklyn-Miller has argued for much more attention to the biomechanics of running to treat CECS.^{62, 63} To reduce load on tibialis anterior, he promotes midfoot landing rather than heel-strike and an increased cadence of 5-10%. Other gait adjustments are made according to individual assessment, such as reducing the vertical tibial angle at foot strike, promoting a smooth gait pattern, a more anterior centre of mass and a shortened stride length.⁶²

A study conducted among military personnel studied the effect of using a forefoot strike pattern among individuals with CECS symptoms for at least 10 months who had an initial hindfoot strike gait pattern.⁶⁴ Six weeks after the implementation of forefoot strike technique, subjects had reduced post-running ICP, reduced pain during running and increased running distance. Improvements were maintained at the 1-year follow-up, where subjects also had an improved 2-mile time trial performance. Most importantly, surgical intervention was not required in any of the individuals.⁶⁴

Another non-operative CECS management strategy is the use of botulinum toxin. It reduces muscle hypertrophy and provide analgesic properties in treating myofascial conditions involving contractures and hypertrophy.⁶⁵ In one case series, intramuscular injection of botulinum



Figure 38.11 Soft tissue therapy in the treatment of anterior compartment syndrome. Sustained myofascial tension with active or passive plantarflexion

toxin resulted in significant reduction of ICP at 1 minute and 5 minutes post exercise in both the anterior and lateral compartments. Subjective exertional pain scores also decreased, with 15 out of 16 study participants reporting no exertional pain at nine-month follow-up. However, the level of evidence to support this treatment is currently low, with several questions remaining regarding the exact mechanism and long-term outcome.⁶⁶

When conservative treatment fails, surgical release is required, especially when there are no obvious precipitating factors to correct. Fasiectomy is rarely necessary for the anterior or lateral compartments, as they release with minimal incision. Indeed, percutaneous and endoscopically assisted releases are approaching 90% success rates. Newer equipment including balloon catheters or endoscopic vein harvesting retractors have been used to make endoscopic-assisted techniques even safer.⁶⁷ Special care is essential to visualise the superficial peroneal nerve at the time of surgery to avoid iatrogenic injury.

Acute compartment syndromes

Acute compartment syndromes are usually associated with trauma. Intracompartmental pressures are significantly elevated and do not subside with rest. Immediate surgical release is essential to avoid ischaemic injury.

It is important to note a number of case reports which describe acute anterior compartment syndromes brought on by exercise rather than trauma.^{68, 69} In many of these cases, the athlete continued to exercise through their initial pain, which was most likely an exertional compartment syndrome that would otherwise have gone away with rest. When pain does not subside within an appropriate time, conversion of an exertional compartment syndrome to an acute compartment syndrome should be considered.

Outcomes of exertional compartment syndrome surgery

As noted previously, the majority of patients undergoing surgery for CECS (80-90%) have a satisfactory result, with many being able to return to their previous level of sport.⁷⁰ However, some patients either fail to improve after surgery or, after a period of improvement, have a recurrence of symptoms. Micheli noted a slightly decreased rate of successful outcomes among female patients,⁷¹ but others have not found any difference in outcome between sexes.⁷² Some studies suggest that failure and recurrences are more common in the deep posterior compartment,^{70, 73} possibly due to failure to release the tibialis posterior compartment.⁷⁰ Indeed, there are few studies looking at outcomes of isolated CECS of the deep posterior compartment.

Patient satisfaction rates for surgical fasciotomy of the deep compartment have been reported to be as low as 52%.⁷⁴⁻⁷⁶ This has been attributed to surgical technique, anatomic variation, over-extensive release (e.g. all four compartments) and inaccurate diagnosis. Confounding factors which may lead to inaccurate diagnosis of posterior CECS include venous insufficiency, popliteal and peroneal entrapment, and tibial stress fracture/periostitis. Thus, accurate diagnosis and conservative management may play a key role in avoiding poor outcomes associated with fasciotomy of the deep posterior compartment.

Several other factors are also associated with surgical outcomes for isolated CECS of the deep compartment. Preoperative ICP measured at four different time intervals (pre, immediately, 1 minute post, 5 minutes post exercise) predict outcomes: a higher area under the 4-point curve is an indicator of likely surgical success.⁷² In addition, a delayed diagnosis is associated with poorer outcomes.⁷²

In a study of 18 patients who underwent revision surgery, increased pressure was found only in a localised area at the site of the scar, whereas 40% had high pressures throughout the compartment.⁷⁵ The exuberant scar tissue was thicker, denser and more constricting than was the original fascia. Eight of the 18 patients had entrapment of the superficial peroneal nerve with numbness and paraesthesia over the dorsum of the foot with exertion, positive Tinel's sign, and localised tenderness over the nerve, exacerbated by active dorsiflexion and eversion, as well as passive inversion and plantar flexion. All those with peroneal nerve entrapment had a good result from the revision surgery, whereas only 50% of those without nerve entrapment had a satisfactory outcome.

Lessons from surgery with unsatisfactory outcomes

The foundation of a successful surgical result begins with a proper anatomic diagnosis. The diagnosis should be confirmed preoperatively with intracompartmental pressure measurement, and all associated or contributing factors should be addressed.

- Surgery should target the specific anatomic pathology and the release of all four compartments should be avoided to reduce the risk of surgical complications.
- Meticulous control of intraoperative bleeding will reduce the risk of postoperative haematoma and cellulitis.
- Perioperative antibiotics and postoperative cryotherapy can reduce the risk of postoperative cellulitis or infection. If identified in the postoperative period, the surgeon should have a relatively low threshold to return to the operating room and perform early irrigation.
- A common complication is postoperative stiffness, which can be avoided by early and aggressive postoperative mobilisation.
- The absolute indication for fasciectomy in contrast to fasciotomy is not clear, as the former may increase the risk of bleeding and postoperative stiffness.

Rehabilitation following compartment syndrome surgery

The following protocol is recommended:⁴

- perioperative antibiotics and cryotherapy to reduce complications of infection, haematoma and cellulitis
- range of motion exercises of the knee and ankle in the immediate postoperative period. Full plantar and dorsiflexion is encouraged
- three to five days of limited weight-bearing on crutches, then full weight-bearing as tolerated

- once wounds have healed, a strengthening program including cycling and swimming to commence
- gradual return to light jogging at about 4-6 weeks after surgery
- full sports participation anticipated at 6-8 weeks (if one compartment released) and 8-12 weeks (if both legs and multiple compartments released)
- athlete should be pain-free with 90% strength regained prior to full sports participation.

LESS COMMON CAUSES

Stress fracture of the fibula

Stress fractures of the fibula are not as common as stress fractures of the tibia. As the fibula plays a minimal role in weight-bearing, this stress fracture is usually due to muscle traction or torsional forces placed through the bone. In the athlete with excessive subtalar pronation, the peroneal muscles are forced to contract harder and longer during toe-off. Examination may reveal local tenderness and pain in springing the fibula proximal to the site of the stress fracture.

This injury is usually not as painful on weight-bearing as is a stress fracture of the tibia. It is treated symptomatically with rest from activity until bony tenderness settles. Due to poorer rotational control, knee-high pneumatic braces may not be as effective as on the tibia. There should then be a gradual increase in the amount of activity. Soft tissue symptoms should be treated. This injury is often associated with a biomechanical abnormality such as excessive pronation or excessive supination.

Referred pain

Referred pain is not a common cause of leg pain in athletes, but should be considered in cases with persistent and atypical pain. Pain may be referred from the lumbar spine, proximal nerve entrapment, the knee joint (Baker's cyst, meniscal cysts), the superior tibiofibular joint (instability or ganglion cyst), and, occasionally, the ankle joint (instability, Maisonneuve fracture).

Nerve entrapments

Within the leg itself, nerve entrapment of either the superficial peroneal nerve in the lateral compartment or the deep peroneal nerve in the anterior compartment can occur due to trauma or a tight brace or cast. Fascial entrapment at the level of the fibular head may also occasionally occur.

The tibial nerve in the deep posterior compartment is less commonly involved with entrapment, but can be injured with trauma. Pain and sensory changes may occur. This diagnosis may be suspected in the presence of motor or sensory changes, and it is confirmed with nerve

conduction studies performed pre- and post-exercise. Surgery may be required to alleviate these conditions.

Vascular entrapments

Popliteal artery entrapment syndrome usually presents with calf pain and is therefore more fully described in Chapter 39, but may rarely present as pain in the anterior compartment.⁷⁷ It can be misdiagnosed as anterior compartment syndrome as they both present with claudication-type pain. However, the pain from popliteal artery entrapment disappears immediately on cessation of exercise, whereas compartment syndrome pain often persists for approximately 30 minutes as an aching sensation. While DVT is most commonly posterior, chronic venous stasis changes can occur anteriorly and may be evidence of systemic disease.

Developmental issues

Juvenile tibia vara (Blount disease) usually presents due to deformity rather than pain. Osgood-Schlatter disease is a traction apophysitis at the insertion of the patellar tendon onto the tibial tuberosity seen commonly among adolescent athletes. Patients usually present with pain and tenderness at the tibial tuberosity (Chapter 44). 'Growing pains' may affect the leg and are usually a diagnosis of exclusion. Intermittent achy pain exacerbated by periods of active growth with completely negative imaging and work-up are characteristic. The youngest reported patient treated with surgical release for pressure positive CECS was 12 years old and it is unclear whether this patient would have grown out of the problem at maturity.

Periosteal contusion

Periosteal contusion occurs as a result of a direct blow from a hard object such as a football boot. It can be extremely painful at the time of injury but the pain usually settles relatively quickly. Persistent pain may occur because of a haematoma having formed under the periosteum. There will be local tenderness and bony swelling. Treatment consists of rest and protection.

Fractured tibia and fibula

Combined fractures of the tibia and fibula are often due to indirect violence in landing from a jump on a twisted foot. The tibia or fibula may be injured individually by a direct blow. Weight-bearing will be impossible in cases of displaced fracture of the tibia. Fracture of the tibia is often compound and visible through damaged skin. An isolated fracture of the fibula may exhibit only local tenderness. An isolated fractured fibula (except where it involves the ankle joint) requires analgesia only. However, the ankle and knee joint must be carefully examined for associated injuries (see Maisonneuve fracture, Chapter 41). The management of a combined fracture is that for fracture of the tibia. Patients with compound (open) fractures should be admitted to hospital.

Many closed tibial fractures can be treated conservatively as long as angulation is minimal. This involves an above-knee plaster with the knee slightly flexed and the ankle in 90° of dorsiflexion. The limb is elevated for 3–7 days until swelling subsides. A check radiograph should be taken after casting. Patients should be reviewed at 6–8 weeks, at which time they may be able to be placed in a hinged knee cast.

Bony union requires 8–12 weeks; however, 16–20 weeks are required for consolidation. Physiotherapy after removal of the plaster is aimed at regaining full range of knee flexion and quadriceps muscle strength. Activities such as swimming can be resumed immediately after removal of the plaster but multidirectional running sports must wait until range of movement and muscle strength have returned to normal.

PRACTICE PEARL

There is a strong trend toward early surgical fixation of unstable tibial fractures with intramedullary nailing, which allows athletes earlier weight-bearing, conditioning activities and return to sport, and obviates the risk of potential mal-union.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e